What does externalisation mean? Four mechanisms, tested with pre-registered evidence

EXCHANGE

writing frees the workspace

test. does a value's internal presence dip once written, until it is reused?

OPEN

design confound found. source text stays readable in context in both modes

DUPLICATION

the page is a redundant backup

test. is a written value still held internally until reused?

OPEN

same confound. no verdict possible from single-context designs

COMPLEMENTARY

page and workspace hold different content

test. are CoT values absent from the direct-mode readout?

REJECTED

values are present, in computation order (rank corr. 0.70, p=2e-8)

EPIPHENOMENAL

CoT helps via extra compute, not content

test. does matched-length filler rescue an ablated model?

REJECTED

no. filler 23% = baseline 19%, real reasoning 98% (n=150)